BIOMETRIC ATTENDANCE DATA REFERENCE

Raw punches, standardized output, and candidate output for records flagged for review

Office shift: 7:00 AM–12:00 PM and 1:00 PM–6:00 PM. Output order is always: Morning IN | Morning OUT | Afternoon IN | Afternoon OUT. "—" means missing. Candidate outputs are not confirmed corrections; records marked for review require verification before being finalized.

Expected output example

Morning IN	Morning OUT	Afternoon IN	Afternoon OUT
6:49 AM	12:10 PM	12:16 PM	6:42 PM

Incorrect raw-data scenarios

No.	Scenario	Example raw punches	Possible output (fixed 4-position format)	Review / rule
01	Duplicate morning IN	IN 6:49:04 AM; IN 6:49:05 AM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM	Earliest valid morning IN; flag duplicate.
02	Duplicate morning OUT	OUT 12:10:24 PM; OUT 12:10:26 PM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM	Duplicate cluster; use configured OUT rule.
03	Duplicate afternoon IN	IN 12:16:07 PM; IN 12:16:08 PM; IN 12:16:10 PM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM	Collapse repeated scans; retain source logs.
04	Duplicate afternoon OUT	OUT 6:42:43 PM; OUT 6:42:47 PM; OUT 6:42:48 PM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM	Flag duplicates; apply configured rule.
05	Missing morning IN	OUT 12:10 PM; IN 12:16 PM; OUT 6:42 PM	— | 12:10 PM | 12:16 PM | 6:42 PM	Missing punch stays blank; don't invent a time.
06	Missing morning OUT	IN 6:49 AM; IN 12:16 PM; OUT 6:42 PM	6:49 AM | — | 12:16 PM | 6:42 PM	Candidate only if 12:16 PM is confirmed afternoon IN; morning OUT remains missing.
07	Missing afternoon IN	IN 6:49 AM; OUT 12:10 PM; OUT 6:42 PM	6:49 AM | 12:10 PM | — | 6:42 PM	Afternoon IN missing; verify.
08	Missing afternoon OUT	IN 6:49 AM; OUT 12:10 PM; IN 12:16 PM	6:49 AM | 12:10 PM | 12:16 PM | —	Afternoon OUT missing; verify.
09	All punches marked IN	IN 6:49 AM; IN 12:10 PM; IN 12:16 PM; IN 6:42 PM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM (candidate)	Candidate assumes chronological punches map to four expected slots; verify actual event types/device setup.
10	All punches marked OUT	OUT 6:49 AM; OUT 12:10 PM; OUT 12:16 PM; OUT 6:42 PM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM (candidate)	Candidate assumes chronological punches map to four expected slots; verify actual event types/device setup.
11	Reversed morning sequence	OUT 6:49 AM; IN 12:10 PM	6:49 AM | 12:10 PM | — | — (candidate)	Candidate uses timestamps as expected slots only; verify labels before accepting.
12	Reversed afternoon sequence	OUT 12:16 PM; IN 6:42 PM	— | — | 12:16 PM | 6:42 PM (candidate)	Candidate uses timestamps as expected slots only; verify labels before accepting.
13	Extra lunch punches	IN 6:49 AM; OUT 12:10 PM; IN 12:12 PM; IN 12:16 PM; OUT 6:42 PM	6:49 AM | 12:10 PM | 12:12 PM | 6:42 PM (candidate)	12:12 PM chosen as earliest afternoon IN only as a candidate; review 12:16 PM duplicate/extra punch.
14	Early arrival	IN 5:30 AM; OUT 12:10 PM; IN 12:16 PM; OUT 6:42 PM	5:30 AM | 12:10 PM | 12:16 PM | 6:42 PM	Apply early-arrival/overtime policy.
15	Late morning IN	IN 7:15 AM; OUT 12:10 PM; IN 12:16 PM; OUT 6:42 PM	7:15 AM | 12:10 PM | 12:16 PM | 6:42 PM	15 minutes late if no grace period.
16	Invalid/missing timestamp	IN with no date or unreadable time	— | — | — | — (invalid punch excluded)	Keep invalid record for review; cannot derive a reliable time.
17	Same-time conflicting punches	IN 6:49 AM; OUT 6:49 AM	6:49 AM | — | — | — (candidate)	Use 6:49 AM only as candidate morning IN; conflicting duplicate type needs review.
18	Out-of-order rows	IN 6:49 AM; OUT 6:42 PM; OUT 12:10 PM; IN 12:16 PM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM	Sort chronologically, then validate.
19	Date mismatch	IN Sep 30 6:49 AM; OUT Oct 1 6:42 PM	Sep 30: 6:49 AM | — | — | —; Oct 1: — | — | — | 6:42 PM	Do not automatically pair events across dates; verify source date.
20	Excessive punches	IN 6:49 AM; OUT 8:00 AM; IN 8:01 AM; OUT 12:10 PM; IN 12:16 PM; OUT 2:00 PM; IN 2:01 PM; OUT 6:42 PM	6:49 AM | 12:10 PM | 12:16 PM | 6:42 PM (candidate)	Candidate uses first morning IN, last morning OUT, first afternoon IN, last afternoon OUT; flag intermediate punches.

How to interpret candidate outputs

Candidate output means the program can display a possible four-slot arrangement for review, but it should not silently save that arrangement as final attendance. Where the raw event types conflict, a supervisor or configured device-specific rule must confirm the result.

Recommended program behavior

Preserve original raw punches. Keep the original device event type and timestamp unchanged; store calculated attendance separately.

Use fixed output positions. Always display Morning IN | Morning OUT | Afternoon IN | Afternoon OUT, using “—” for missing values.

Make ambiguity visible. Add a review flag/reason alongside the candidate output. A timestamp candidate is not proof that the event was an IN or OUT.

Use configurable rules. Define duplicate intervals, shift windows, lunch-break handling, grace periods, and manual correction permissions.

Keep an audit trail. Record the original value, accepted correction, reason, editor, and correction time.

Suggested test checklist

No.	Test scenario	Pass condition
1	Normal four-punch day	Output is consistent; ambiguity is flagged; original logs remain unchanged.
2	Duplicate punches within configured threshold	Output is consistent; ambiguity is flagged; original logs remain unchanged.
3	Each missing-punch position	Output is consistent; ambiguity is flagged; original logs remain unchanged.
4	All events marked IN / all marked OUT	Output is consistent; ambiguity is flagged; original logs remain unchanged.
5	Out-of-order raw rows	Output is consistent; ambiguity is flagged; original logs remain unchanged.
6	Punches at shift boundaries	Output is consistent; ambiguity is flagged; original logs remain unchanged.
7	Punches during lunch	Output is consistent; ambiguity is flagged; original logs remain unchanged.
8	Same timestamp with conflicting types	Output is consistent; ambiguity is flagged; original logs remain unchanged.
9	Invalid timestamp or wrong date	Output is consistent; ambiguity is flagged; original logs remain unchanged.
10	Duplicate imports	Output is consistent; ambiguity is flagged; original logs remain unchanged.
11	Manual correction and audit history	Output is consistent; ambiguity is flagged; original logs remain unchanged.